

Day 1 – AI Foundations: What AI Really Is (Full Teaching Presentation)

This document is a **complete Day 1 presentation** meant to be used directly for teaching.
You can copy each slide section into PowerPoint / Google Slides as-is.
No focus on visuals or cosmetics — this is **content-first**.

Slide 1 — Title

AI Foundations

What AI Really Is (And What It Is Not)

Purpose of today: - Reset mental models - Remove hype - Build correct AI thinking from day one

Slide 2 — Why Are We Even Doing This?

You already: - Know how to call AI / LLM APIs - Understand basic NLP terminology - Can integrate AI into products

Yet: - AI requirements feel unclear - Decisions feel guessy - Interviews expose gaps

Reason: The problem is not coding. The problem is understanding.

Slide 3 — The Real Gap We Are Fixing

Most engineers struggle with: - Identifying whether a problem is an AI problem - Knowing *which type* of AI to use - Explaining AI choices to management - Defending AI decisions in interviews

This program fixes **thinking**, not syntax.

Slide 4 — What This Program Is NOT

We are NOT: - Turning you into mathematicians - Teaching heavy statistics - Focusing on frameworks or vendors - Making you data scientists

We ARE: - Teaching AI decision-making - Teaching AI system thinking - Teaching how to explain AI clearly

Slide 5 — What Is Artificial Intelligence?

Artificial Intelligence is the ability of a system to: - Learn patterns from data - Make predictions or decisions - Improve performance with experience

Key idea: AI learns correlations, not understanding.

Slide 6 — AI Is Not Human Intelligence

AI is NOT: - Conscious - Self-aware - Reasoning like humans

AI IS: - Large-scale statistical pattern recognition - Optimization over examples

Think autocomplete — but at massive scale.

Slide 7 — AI vs Machine Learning vs Deep Learning

- **Artificial Intelligence:** Broad goal
- **Machine Learning:** Learning from data
- **Deep Learning:** Neural networks with many layers

Hierarchy: AI → ML → Deep Learning

Slide 8 — Where Do LLMs Fit In?

- LLMs are a **type of deep learning model**
- Trained on massive text datasets
- Specialized for language tasks

LLMs are powerful — but limited to what they were trained on.

Slide 9 — Why LLMs Feel Like Real Intelligence

LLMs feel intelligent because they: - Speak fluently - Adapt across tasks - Mimic reasoning patterns

But they do not *understand* — they predict the next token.

Slide 10 — The Danger of LLM-First Thinking

Common mistakes: - Using LLMs for simple classification - Ignoring structured data - Paying high costs for low-value tasks

Not every problem needs an LLM.

Slide 11 — What AI Is Actually Good At

AI excels at: - Pattern detection - Prediction - Classification - Automation at scale

AI shines when rules become too complex.

Slide 12 — What AI Is Bad At

AI struggles with: - No or poor-quality data - Rapidly changing logic - Rare edge cases - True reasoning and judgment

AI fails outside its training distribution.

Slide 13 — Example: Spam Detection

Input: - Email text

Output: - Spam / Not spam

Type: - Classification

No LLM required.

Slide 14 — Example: Price Prediction

Input: - Historical prices - Market signals

Output: - Numeric value

Type: - Regression

This is AI — without deep learning.

Slide 15 — Why Engineers Overestimate AI

Reasons: - Demos hide failure modes - Marketing exaggerates success - Costs are not discussed

Demo AI $\neq$ Production AI

Slide 16 — AI Systems vs Traditional Software

Traditional Software: - Rule-based - Deterministic - Predictable failures

AI Systems: - Data-driven - Probabilistic - Uncertain outcomes

AI systems fail differently.

Slide 17 — The Most Important Mental Shift

AI engineering is decision-making, not coding.

If you choose the wrong problem or approach, no model can save you.

Slide 18 — Group Exercise

For each example: 1. Is this an AI problem? 2. Why or why not?

Examples: - Password validation - Fraud detection - Resume screening - FAQ chatbot

Slide 19 — Key Takeaways

- AI learns patterns from data
- LLMs are a subset of AI
- Most AI failures happen before coding

Clarity beats cleverness.

Slide 20 — What Comes Next

Next sessions will cover: - How machine learning actually learns - Classification vs regression - How to identify AI problems correctly

Slide 21 — Reflection / Homework

Before next session: - Watch: StatQuest – What is Machine Learning? - Think: Where AI should NOT be used in your current work

End of Day 1 Presentation